Decision Making Analysis

Prototype Sophistication for a Trapped-Ion Qubit Control Module

A Design Engineer's Perspective

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EECE 102 - Transformational Technology Case Study

Howard University

April 2026

1. Overview

This case study takes the perspective of a design engineer at a mid-stage quantum computing startup that has raised a Series A round and is racing to ship its first demonstration system. The company uses trapped-ion qubit technology and competes indirectly with Quantinuum and IonQ. My specific job is to lead the hardware design of the qubit control module, the mixed-signal subsystem that generates the RF and laser-pulse waveforms that manipulate each ion's quantum state.

The central question: how sophisticated should the first prototype be? Three concrete options sit on the table, each representing a different bet on scope, cost, schedule, and technical risk:

- Option A - Minimal Prototype: 16 qubits, off-the-shelf FPGA, development cost ~$450K, 6-month timeline.
- Option B - Mid-Tier Prototype: 32 qubits, custom FPGA with partial ASIC integration, cost ~$1.1M, 12-month timeline.
- Option C - Full-Spec Prototype: 50 qubits, fully custom ASIC control, cost ~$2.8M, 20-month timeline.

The nature of the decision is both technical and strategic. Technically, a more sophisticated prototype lets us demonstrate higher qubit counts and tighter gate fidelities, which is what investors and early customers want to see. Strategically though, the startup has roughly 18 months of runway and a small team, so a prototype that slips or fails to function burns cash we cannot replace. This is exactly the kind of decision where outcomes are not mediocre, but tend to be extreme in either direction.

Why Outcomes Are Extreme, Not Mediocre

Quantum hardware is a field where partial success translates poorly. A control module that hits 90% of its gate-fidelity target is not 90% as valuable as one that hits 100%; it is often unusable for the error-correction thresholds customers actually care about. This creates a sharp cliff in the value function. Option A produces a modest but reliable demo. Option C, if it works, positions the company as a technical leader and likely closes a Series B at a strong valuation; if it fails, it consumes most of the runway and effectively ends the company. Option B sits in between but is not a simple average — the mid-tier design preserves optionality, letting us upgrade to a full ASIC later once we have de-risked the core blocks.

2. Sources of Uncertainty and Risk Mitigation

Every input to this decision is uncertain. The most important sources are:

- Technical yield of the custom silicon. First-pass ASIC success rates for mixed-signal designs at this complexity are historically below 50%.

- Laser and trap infrastructure readiness. The control module cannot be tested in isolation; it depends on the ion-trap hardware being ready on the same timeline.
- Talent availability. McKinsey reports only one qualified quantum candidate exists for every three open quantum job openings [1], and industry forecasts project a need for roughly 250,000 quantum professionals by 2030 [2]. This matches what I already see in practice: hiring a senior RF/mixed-signal engineer in 2026 is painfully slow.
- Market timing. If Quantinuum or IonQ ships a competing system first, the value of our prototype drops sharply regardless of its technical quality. Quantinuum's November 2025 Helios launch: 98 physical qubits, 48 logical qubits, two-qubit gate fidelity above 99.9%, sets the current bar for trapped-ion systems [3].
- Black swan: A post-quantum cryptography regulation wave that redirects customer funding away from hardware and toward software/PQC migration. The McKinsey 2025 monitor already flags PQC transition lag as a top-three executive risk [4].

Mitigation Steps

To guard against these risks I would build in several redundancies and real options:

- Redundant IP paths. License a backup FPGA IP core for the pulse generator so that a failed custom block does not kill the build.
- Staged gate reviews. Commit budget in tranches tied to technical milestones (waveform generator, control loop, end-to-end fidelity) rather than up-front.
- Option to upgrade. Choose architectures where the FPGA prototype shares pinout and firmware hooks with the eventual ASIC, so the mid-tier design is a real stepping stone, not a throwaway.
- Parallel vendor qualification. Pre-qualify two foundries early so we are not hostage to a single fab's schedule.
- Insurance against black swans. Maintain a 3-month cash reserve outside the prototype budget, and keep one engineer allocated to a lightweight software offering that could generate bridge revenue.

These steps do not eliminate risk, but they convert a single binary bet into a sequence of smaller bets that can be cancelled or upgraded as new information arrives.

3. Decision Tree Analysis

The flowchart below captures the full decision space. Each branch carries a probability estimate and a net dollar outcome in millions (value of demonstrated capability minus the development cost). Probabilities are my engineering estimates, informed by 2025 industry data on trapped-ion gate-fidelity benchmarks [3], the general quantum hardware milestone slippage reported across the McKinsey 2025 monitor [4], and my own course case study on quantum computing as a transformational technology [5].

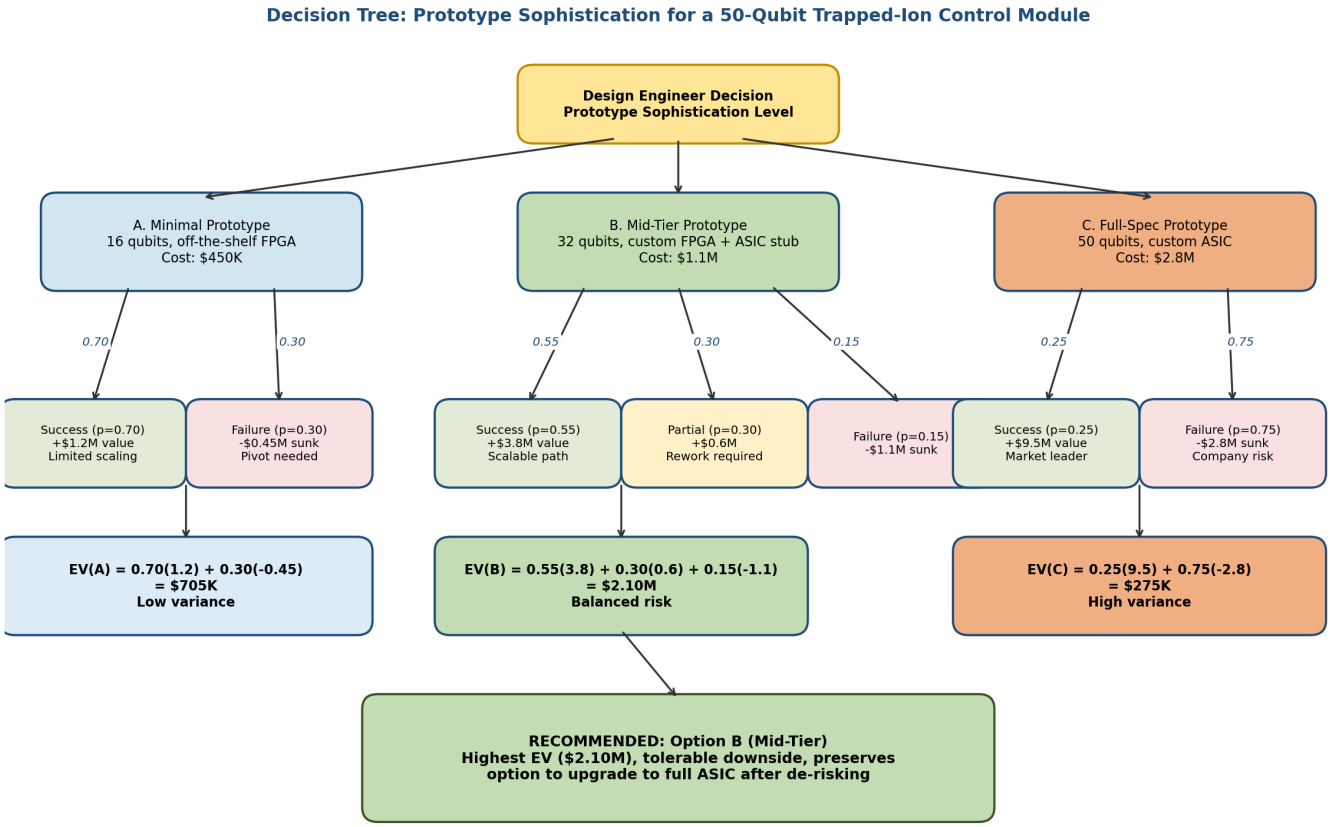

Figure 1. Decision tree with probabilities, outcomes, and expected values.

Expected Value Summary

Option	EV ($M)	Upside	Downside	Std Dev
A — Minimal	0.71	+1.20	-0.45	0.76
B — Mid-Tier	**2.11**	+3.80	-1.10	1.95
C — Full-Spec	0.28	+9.50	-2.80	5.33

Option B dominates on expected value with a 2.11 million dollar EV, roughly triple Option A and nearly eight times Option C. Option C has the highest upside by far, but the 75% failure probability drags its EV near zero and its standard deviation is over five million dollars, meaning the downside in any single trial is catastrophic for a company with 18 months of runway.

4. Cost Function and Opportunity Cost

Expected value alone does not capture everything that matters. I built a five-attribute cost function that weighs financial EV alongside technical risk, schedule risk, strategic fit, and team readiness. Each option is scored 1 to 10 on each attribute, then combined with weights that sum to 1.0. Higher composite scores are better.

Default weights: financial EV 30%, technical risk 25%, schedule risk 15%, strategic fit 20%, team readiness 10%. The full implementation is in the accompanying Python script quantum_decision.py.

Composite Scores

Option	Composite Score	Opportunity Cost	Rank
A — Minimal	6.50	0.50	2
B — Mid-Tier	**7.00**	**0.00**	**1**
C — Full-Spec	4.60	2.40	3

Opportunity cost is the gap between each option's score and the top-ranked option. Choosing Option A instead of B costs us 0.50 composite points, mostly on financial EV and strategic fit. Choosing Option C costs 2.40 points, the penalty concentrated on technical risk, schedule risk, and team readiness, since our team has not previously taped out a custom ASIC at this complexity.

Sensitivity Analysis

I stress-tested the recommendation by sweeping two parameters:

- Option B's success probability from 0.35 to 0.75. Even at the pessimistic 0.35 success rate, EV remains positive at $1.35M; still well above Option C and comparable to Option A.
- The financial EV weight from 15% to 45%. The composite score for Option B shifts only from 6.79 to 7.21, so the ranking is stable across any reasonable weighting.

The conclusion is robust: Option B wins across a wide band of assumptions. The only scenario where Option C becomes attractive is if the success probability for the full ASIC rises above ~45%, which would require either hiring a second ASIC lead or partnering with an experienced foundry, both of which are themselves multi-month decisions.

Code: Cost Function Implementation

The core of the analysis is a weighted composite score. Each option is rated 1 to 10 on five attributes and combined with weights that sum to 1.0. Opportunity cost is the gap between any option and the top-ranked alternative.

```
ATTRIBUTES = ["financial_ev", "technical_risk", "schedule_risk",
              "strategic_fit", "team_readiness"]

SCORES = {
    "A_Minimal":  {"financial_ev": 4, "technical_risk": 9,
                   "schedule_risk": 9, "strategic_fit": 4, "team_readiness": 9},
    "B_MidTier":  {"financial_ev": 8, "technical_risk": 6,
                   "schedule_risk": 6, "strategic_fit": 8, "team_readiness": 6},
    "C_FullSpec": {"financial_ev": 5, "technical_risk": 2,
                   "schedule_risk": 2, "strategic_fit": 10, "team_readiness": 3},
}

DEFAULT_WEIGHTS = {
    "financial_ev":   0.30,
    "technical_risk": 0.25,
    "schedule_risk":  0.15,
    "strategic_fit":  0.20,
    "team_readiness": 0.10,
}

def cost_score(option, weights=None):
    """Weighted composite score. Higher is better."""
    w = weights or DEFAULT_WEIGHTS
    s = SCORES[option]
    return sum(w[a] * s[a] for a in ATTRIBUTES)

def opportunity_cost(option, weights=None):
    """Gap between this option and the best alternative composite score."""
    best = max(cost_score(o, weights) for o in TREE)
    return best - cost_score(option, weights)
```

Worked Calculation

Plugging Option B (Mid-Tier) into cost_score by hand shows exactly how the weights combine the five attribute scores into a single number:

```
cost_score("B_MidTier")
  = 0.30 * 8   # financial_ev
  + 0.25 * 6   # technical_risk
  + 0.15 * 6   # schedule_risk
  + 0.20 * 8   # strategic_fit
  + 0.10 * 6   # team_readiness
  = 2.40 + 1.50 + 0.90 + 1.60 + 0.60
  = 7.00
```

The same calculation for the other two options yields 6.50 for Option A and 4.60 for Option C. Opportunity cost then subtracts each option's score from the maximum (7.00), giving 0.50 for A, 0.00 for B, and 2.40 for C. These are the values that appear in the table above.

Sample Program Output

Running the full script produces the following output. The bottom section confirms the recommendation by picking the option with the highest expected value and composite score.

```
====================================================================
  EXPECTED VALUE ANALYSIS
====================================================================
Option        EV ($M)     Upside      Downside    Std Dev
-------------------------------------------------------------
A_Minimal     0.705       1.200       -0.450      0.756
B_MidTier     2.105       3.800       -1.100      1.949
C_FullSpec    0.275       9.500       -2.800      5.326

====================================================================
  COST FUNCTION (Composite Score, higher = better)
====================================================================
Option        Score     Opp. Cost
---------------------------------------
A_Minimal     6.50      0.50
B_MidTier     7.00      0.00
C_FullSpec    4.60      2.40

====================================================================
  SENSITIVITY: Success probability shift on Option B
====================================================================
Delta     New P(Success)    New EV ($M)
---------------------------------------------
-0.20     0.35              1.352
-0.10     0.45              1.728
+0.00     0.55              2.105
+0.10     0.65              2.482
+0.20     0.75              2.858

====================================================================
  SENSITIVITY: Financial weight shift on Option B score
====================================================================
Delta     New Weight    New Score
---------------------------------------
-0.15     0.15          6.79
-0.05     0.25          6.93
+0.00     0.30          7.00
+0.05     0.35          7.07
+0.15     0.45          7.21

====================================================================
  RECOMMENDATION
====================================================================
Recommended option: B_MidTier
```

```
Expected value:   $2.10M
Composite score:  7.00/10
Opportunity cost: 0.00
```

Two things stand out in the output. First, the expected value table makes the variance gap visible: Option C has 7x the standard deviation of Option A, which is the quantitative version of saying it is a company-ending bet if it fails. Second, the sensitivity sweeps show the recommendation is not fragile: Option B's composite score stays in a narrow 6.79 to 7.21 band across a ±15 point shift in the financial weight, and its EV stays comfortably positive even if its success probability is 20 points lower than my baseline estimate.

5. Recommendation

My recommendation as design engineer is to build Option B: a 32-qubit mid-tier prototype using a custom FPGA with a partial ASIC integration stub. The reasoning is straightforward:

- Expected value of $2.11M is roughly 3x Option A and 8x Option C.
- Composite score of 7.0 leads the field with zero opportunity cost under default weights.
- Downside of $1.1M is painful but survivable within the 18-month runway, unlike Option C's $2.8M loss which would trigger layoffs.
- The architecture preserves a real option to upgrade to a full ASIC in the next design cycle, reusing the firmware and control-loop IP we develop now.
- It matches current team readiness. We have the FPGA expertise in house; we do not yet have a proven full-ASIC tapeout team.

To implement this recommendation well, I would pair the Option B build with three supporting commitments. First, structure the budget in three tranches gated on waveform generation, closed-loop control, and end-to-end fidelity, so that the board can pull the plug cleanly if any gate fails. Second, begin hiring one senior ASIC engineer immediately so that if Option B succeeds, we are ready to start a full-spec follow-on within a quarter. Third, publish a short technical note on the control architecture — in line with the broader 2025 trend of quantum hardware companies using openness to attract talent and credibility, as seen in IBM's published quantum roadmap [6] and Quantinuum's Helios announcement and $800M raise at a $10 billion valuation [3, 7].

The deeper lesson from running this analysis is that in quantum hardware, the temptation to reach for the most ambitious prototype is strong because the upside is genuinely transformational. But at an early stage, with a small team and finite runway, the best move is usually the one that keeps you in the game long enough to make the ambitious bet later from a stronger position. Option B does exactly that.

6. Additional Credit: Interactive Decision Program

The accompanying script quantum_decision.py includes an interactive() function that lets a user enter their own weights for the five attributes. The program normalizes them, recomputes composite scores and expected values, ranks all three options, and prints a recommendation based on the user's priorities. This makes the analysis reusable for other quantum hardware decisions: for example, a government fund manager with a stronger weight on strategic fit and a smaller weight on schedule risk would see Option C move up the ranking, which matches the intuition that different decision makers should reach different conclusions from the same underlying numbers.

The script can be extended with an LLM backend (the Anthropic API, for example) to let a user describe a decision in natural language and have the model extract options, probabilities, and weights automatically. This would turn the analysis from a one-off case study into a general-purpose decision support tool.

References

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